

Learning-Rate Warmup and Pruning Robustness in an MLP for Protein Structure Quality Prediction

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1 Introduction

This report trains multilayer perceptrons (MLPs) on the *Physicochemical Properties of Protein Tertiary Structure* dataset [4]. Each row is a candidate 3D structure (a decoy) of a protein, described by nine features (F1-F9). The target is its RMSD, the deviation from the true structure in ångströms ($\text{\AA}$; $1 \text{\AA} = 0.1 \text{ nm}$). This is a regression task which estimates how wrong a decoy is without knowing the true structure. Three MLP design axes are studied first (optimiser, architecture, activation function), followed by a test of whether learning-rate warmup improves robustness to magnitude-based pruning, and whether it does so by enabling a larger learning rate (LR), denoted η . For every experiment, a hypothesis was made and the experiment itself either supported or contradicted the claim. All numbers are reproducible from the accompanying notebook.

2 Data preparation

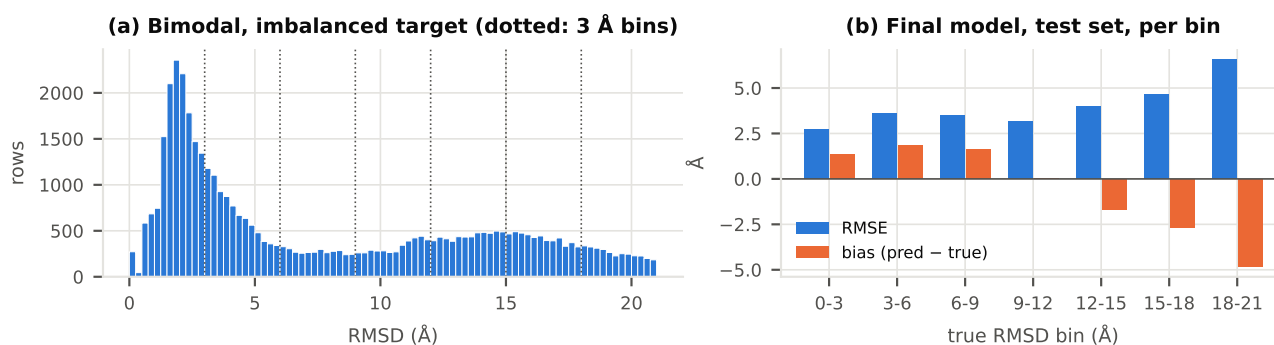


Figure 1: (a) The target is bimodal: a near-native peak at 1-3 Å and a broad hump at 11-17 Å. (b) The final model’s test error and bias (prediction – truth) per RMSD bin.

Cleaning: 1,711 exact duplicate rows (3.7%) were removed before splitting, leaving 44,019 rows. A duplicate on both sides of a split would leak training rows into evaluation.

Feature relevance: Linear correlation with RMSD is near zero for most features, yet every feature’s mutual information (MI, 0.18-0.35) is far above that of random-noise. F7, for example, has $r = -0.003$ but the highest MI. The relationships are non-linear, so a correlation filter would wrongly discard useful features. Two features are nearly redundant. $\mathbf{F2} = \mathbf{F1} \times \mathbf{F3}$ exactly (non-polar exposed area = total area $\times$ non-polar fraction), so it was dropped. $\mathbf{F5} \approx 138.5 \times \mathbf{F1}$ was dropped at first too, but its small deviation has MI 0.089. An MLP with F5 beat one without it in 5/5 seeds (CV RMSE 4.10 vs 4.16 Å), so F5 was restored. This gives **8 inputs**.

Outliers: Seven features are right-skewed (skew > 1). On such data the IQR rule mostly flags the natural long tail. After $\log_{1p}$, almost all flags vanish except for F7, whose maximum is $\sim 28\times$ its median. Those rows have ordinary RMSD values, so they were kept. Instead, every feature is log-transformed (if skewed) and clipped to its training 0.5th-99.5th percentile. The target is never clipped, because a large RMSD is a genuine bad decoy.

Scaling: Features are standardised (z-scores), and so is target for training. Metrics are reported in Å. Z-scores were preferred to min-max scaling because they are zero-centred (all-positive inputs force a neuron’s weight gradients to share a sign), they match the unit-variance assumption of Xavier/He initialisation, and they keep sigmoid/tanh out of saturation. Every transform is fitted on training data only.

Splitting: An 80/20 train/test split (35,215 / 8,804 rows), stratified on 3 Å RMSD bins, with a fixed seed. The test set was used only for the final results. All model comparisons use 5-fold stratified cross-validation (CV) on the training set, and 10% of each fold’s training part is held out for early stopping, so the stopping epoch is never chosen on scored data. A relevant limitation here is that since there are no protein IDs, decoys of the same protein can appear on both sides of a split.

Imbalance: The 0-3 Å bin holds 34% of rows and the 18-21 Å bin 7% (Fig. 1a). It is handled through stratified folds and test split, and through per-bin RMSE and bias at evaluation. Inverse-bin-frequency loss weights were tested and rejected. They cut rare-bin error by up to 0.47 Å but added 1.0 Å in the common 0-3 Å bin, and overall

RMSE rose from 4.09 to 4.31 Å. Nor did they remove the under-prediction of high RMSD, which stems from the features, not the loss.

3 Core investigation

Baseline: A 2×64 ReLU MLP (He initialisation, Adam with $\eta = 0.001$, batch 256, early stopping with patience 20) reaches CV RMSE **4.09 Å** ($R^2 = 0.555$). For comparison: predicting the mean gives 6.14 Å, linear regression 5.19 Å ($R^2 = 0.29$), and gradient-boosted trees 4.13 Å. Each axis below changes one setting, with 3 seeds $\times$ 5 folds per configuration. Convergence speed is the number of epochs until the early-stopping MSE (on standardised RMSD, where predicting the mean scores 1.0) drops below 0.50.

Optimisers (Fig. 2a): Momentum gave a $\sim 20\times$ speed-up over plain SGD and beat RMSprop, so much of Adam’s speed seems to come from the momentum it contains. Final quality was *not* equal (A2 contradicted). Under a fixed budget and early stopping, plain SGD never reached the good region: at small η its best epoch was the 500-epoch limit. RMSprop was the most fragile optimiser at high η (it diverged at $\eta \geq 0.03$). The reason is that RMSprop scales each step by a running average of past gradient sizes, and that average starts at zero. Unlike Adam, PyTorch’s RMSprop does not correct for this (no *bias correction*), so its very first step is $10\times$ larger than η . Oversized early steps are exactly the problem warmup addresses. Large η also killed ReLU units (Adam: 0.3% dead at $\eta = 0.001$, 48% at $\eta = 0.1$).

Architecture (Fig. 2b): Every increase in width or depth lowered RMSE, consistently across seeds. Bigger neural networks overfit sooner (larger train-validation gap), but early stopping halted them first, so the largest neural network was best. Depth was more parameter-efficient: 4×64 (13k weights) matched 2×256 (68k). Hypotheses B1/B2 under-estimated capacity because they assumed the features set a ceiling rather than the model, as suggested by MLP and trees both stopping at $R^2 \approx 0.55$. That ceiling was really the baseline’s limited capacity.

Activations (Fig. 2c,d): Sigmoid’s gradients shrink by $\approx 4\times$ per layer going backwards, matching its maximum derivative of 0.25, and reach $\sim 10^{-6}$ of the output layer’s after 8 layers. tanh and ReLU keep gradients flat. With Adam, which rescales each weight’s step by its own gradient scale, this shows up as slow training (340 vs 8.6 epochs) rather than failure.

Final test evaluation: The best CV-validated configuration (4×256 , ReLU, Adam, $\eta = 0.001$) reaches **test RMSE 3.78 ± 0.03 Å**, $R^2 = 0.621$ (5 seeds), in close agreement with its CV score (3.81 Å), so model selection did not overfit the validation folds. Per bin (Fig. 1b), errors grow towards the rare high-RMSD bins, with regression to the mean: over-prediction of +1.3 Å for 0-3 Å and under-prediction of -4.8 Å for 18-21 Å.

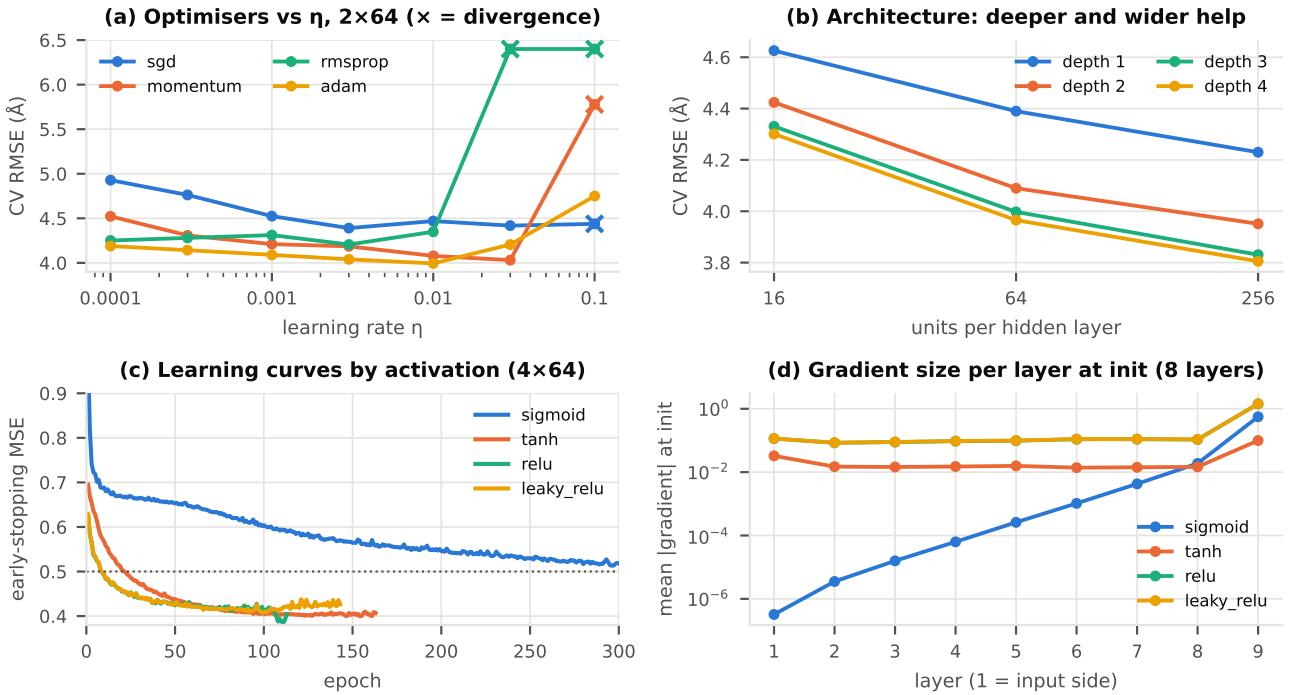


Figure 2: Core investigation (CV on the training set, mean of 3 seeds). (a) Each optimiser across a shared grid of η . (b) RMSE by depth and width. (c) Early-stopping learning curves by activation. (d) Mean gradient magnitude per layer at initialisation, 8 hidden layers. ReLU and Leaky ReLU overlap.

Hypothesis (stated before running)	Verdict and evidence
A1: Adam/RMSprop converge fastest, SGD slowest	Partly. Epochs to 0.50: Adam 9.7, momentum 12.5, RMSprop 29.5, SGD 293
A2: at each one’s best η , all optimisers within 0.05 Å	Contradicted. Adam 3.99, momentum 4.03, RMSprop 4.21, SGD 4.39 Å
A3: adaptive methods tolerate a wider range of η ; momentum fails before SGD	Mixed. Adam most tolerant, RMSprop <i>least</i> ; momentum fails at $\eta = 0.1$, SGD does not
B1: width 16→64 helps, 64→256 <0.05 Å	Partly. −0.33 then −0.14 Å
B2: 2 layers beat 1; more adds <0.05 Å	Half. 1→2: −0.30 Å; 2→4: −0.12 Å
B3: big nets overfit sooner; early stopping protects them	Supported. Best epoch 310→54; best net is 4×256 (3.81 Å)
C1: speed ReLU > tanh > sigmoid	Supported. 8.6 / 21 / 340 epochs (4×64)
C2: sigmoid gradients vanish with depth (softened by Adam)	Supported. $\approx 4\times$ shrink per layer; affects speed more than final RMSE
C3: tanh $\approx$ ReLU $\approx$ Leaky ReLU	Partly. Leaky = ReLU (<0.5% dead units); tanh beat ReLU at depth 4 (3.90 vs 3.97 Å)

Table 1: Core-investigation hypotheses and verdicts. A difference counted only if it exceeded seed/fold noise ($\sim 0.015/0.04$ Å) in all seeds.

4 Research investigation: warmup and pruning robustness

Design: The network is 4×256 ReLU, trained with SGD with momentum 0.9 (batch 256) for a fixed 100 epochs with cosine decay. Warmup adds a linear ramp over the first 5 epochs. Choices are made on a 10% validation split, and all pruning results are reported on the test set, over 5 seeds. **Step 1:** train with and without warmup over a grid of peak η (0.005-0.2). **Step 2:** pre-registered rules choose **A** (no warmup, at its best η_0), **B** (warmup, at the largest $\eta_1 > \eta_0$ whose validation RMSE is within 0.05 Å of A’s: comparable unpruned accuracy) and the control **C** (warmup at η_0). **Step 3:** one-shot global magnitude pruning of all weight matrices at 0-98% sparsity, followed by identical fine-tuning ($\eta = 0.001$, 10 epochs, pruned weights held at 0 by masks). Robustness is $\Delta(s) =$ test RMSE after fine-tuning minus unpruned test RMSE. **Hypotheses:** **H1:** warmup raises the maximum stable η . **H2:** B has a smaller Δ than A at sparsities $\geq 70\%$. **H3:** the benefit comes from the larger η , so B beats C while $C \approx A$.

H1: supported. Without warmup, the largest stable η was 0.005. At $\eta = 0.01$, 2/3 seeds diverged, and at $\eta \geq 0.02$ all did, every time within the first epoch. With warmup all runs trained, up to 0.2 (Fig. 3a), a $\geq 20\times$ increase. This is consistent with Kalra and Barkeshli [3]: small early steps move the network to a better-conditioned region, where large steps are safe. The rules selected $\eta_0 = 0.005$ and, for B, $\eta_1 = 0.2$ (validation RMSE 3.73 vs 3.74 Å). Warmup at $\eta = 0.01$ -0.03 was ~ 0.1 Å better than A and therefore not “comparable”.

H2 and H3: supported (Table 2, Fig. 3b,c). At matched unpruned accuracy (B’s test RMSE was even slightly worse), B lost 0.22-0.27 Å less than A, in 5/5 seeds at every sparsity $\geq 70\%$. Right after pruning the gap is dramatic: at 70% sparsity B’s RMSE is 4.6 Å, while A and C jump to 8.0 and 7.4 Å, worse than predicting the mean. The control is decisive. Warmup at the small η (C) brought no benefit, so warmup helps only through the larger η it enables. Alternative explanations were checked: B had the fewest dead units, so its pruning was not “free” sparsity. The fine-tuning η (0.001) was relatively larger for A and C, which should favour their recovery [2]. And measuring Δ against the fine-tuned dense network gives the same ranking.

Optional extensions and further analysis:

- *Warmup duration* (an optional extension in the spec). Keeping B’s $\eta = 0.2$, one epoch of warmup was already enough for stable training. A 15-epoch warmup was *less* robust than a 5-epoch one (4-5/5 seeds), plausibly because it leaves fewer epochs at the large η .
- *Retraining with LR rewinding [2] instead of fine-tuning.* Each pruned network was retrained with its own schedule (its own peak η) rather than at $\eta = 0.001$. Recovery improved for all conditions, but the A-B gap *widened* (B lost only 0.03 Å at 90% sparsity vs 0.58 Å for A), contrary to the prediction. The large η helps twice: it gives more robust weights and more effective retraining.
- *Robustness across all warmup η* (exploratory, not pre-registered; Fig. 3d). H3 rests on two values of η , so all seven warmup runs from Step 1 were pruned and fine-tuned. Robustness rises steadily with η : $\Delta_{70\%}$ falls from 0.35 to 0.09 Å (Spearman $\rho = -0.88$ over 35 runs), even though $\eta = 0.01$ gives the best unpruned model. Robustness tracks η , not the starting accuracy.
- *Post-hoc look at the weights.* B concentrates more of its weight mass in its largest weights (the top 10% hold 60.6% of $\sum w^2$ vs 56.5% for A), one possible reason large- η networks lose less to magnitude pruning.

Interpretation and limitations: The effect reported by Frankle and Carbin [1] was observed, and the evidence supports the proposed mechanism on both links. Warmup raises the tolerable η (H1), and the larger η , not warmup itself, produces the robustness (H3 and the sweep over all warmup η). Warmup’s effect on pruning robustness is therefore *indirect*. Limitations: a single tabular dataset with an MLP (the original studies used CNNs on images),

one-shot rather than iterative pruning, a single fit/validation split, and B’s η lying at the edge of the grid. Large- η training without warmup cannot be tested, because it diverges. Finally, the reason large- η networks are robust (weight concentration or flatter minima) was explored only post hoc.

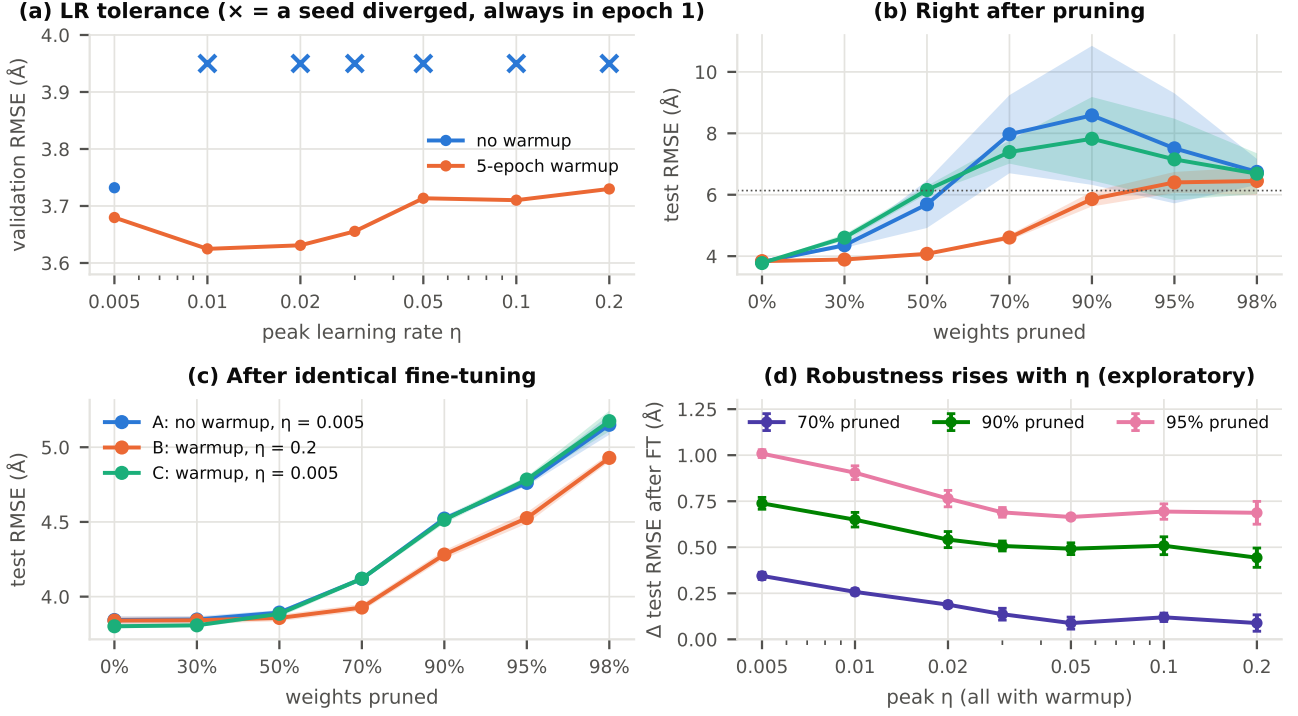


Figure 3: Warmup and pruning (mean $\pm$ std over 5 seeds; (b)-(d) on the test set). (a) Validation RMSE vs peak η ; $\times$ = at least one seed diverged. (b, c) Accuracy vs sparsity for conditions A, B, C, before and after identical fine-tuning (dotted: predicting the mean). (d) Exploratory: Δ vs peak η among warmup-trained networks.

	Unpruned test RMSE	Δ 70%	Δ 90%	Δ 95%	Δ 98%	Dead units
A: no warmup, $\eta = 0.005$	3.81	0.31	0.71	0.95	1.34	3.6%
B: warmup, $\eta = 0.2$	3.84	0.09	0.44	0.69	1.09	0.2%
C: warmup, $\eta = 0.005$	3.78	0.35	0.74	1.01	1.40	0.0%

Table 2: Error added by pruning after fine-tuning (Δ , Å; mean of 5 seeds). B vs A: -0.22 to -0.27 Å and B vs C: -0.26 to -0.32 Å, smaller in 5/5 seeds at every sparsity. C vs A: $+0.03$ to $+0.06$ Å, with C better in only 0-2/5 seeds.

5 Conclusion

Careful data preparation (log transform, clipping, stratification, per-bin evaluation, and checking feature decisions with the model itself) gave a 4×256 MLP with test RMSE 3.78 Å ($R^2 = 0.62$). Capacity, optimiser choice and LR mattered more than the initial hypotheses expected, and sigmoid’s vanishing gradients appeared exactly as theory predicts. In the research investigation, warmup raised the tolerable LR at least 20-fold. At matched accuracy, the warmup-trained network was clearly more robust to magnitude pruning, but only because of the larger LR that warmup unlocks.

References

- [1] J. Frankle and M. Carbin, “The lottery ticket hypothesis: Finding sparse, trainable neural networks,” in *Proc. ICLR*, 2019.
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- [3] D. S. Kalra and M. Barkeshli, “Why warmup the learning rate? Underlying mechanisms and improvements,” in *Advances in NeurIPS*, 2024.
- [4] “Physicochemical Properties of Protein Tertiary Structure,” UCI Machine Learning Repository, dataset 265. <https://archive.ics.uci.edu/dataset/265>.